

Risk II: Comparing Risks, Valuing the Contract

Advanced Microeconomics 1: Economic Decision Making. Lecture 8

Luke Heath Milsom

Where we left the question

- Monday built the machinery: expected utility, risk aversion as concave u , and the price of a risk — its certainty equivalent and premium.
- Two things remain. **(a)** We can price a *given* lottery, but not yet say when one distribution is *better* or *riskier* than another — needed, because the poor and the rich face *different* inflation lotteries.
- **(b)** The contract is still unvalued: indexation transforms the worker's lottery — less price risk, more job risk, from L6. Comparing transformed lotteries is exactly today's first tool.

Today's route

1. Comparing distributions: first- and second-order stochastic dominance.
2. One more tool: updating beliefs (Bayes).
3. The ex ante cost of inflation, by household.
4. The insurance value of indexation, against its price.
5. When the assumptions are wrong — and what that means for our answers.
6. Conclusion.

Comparing distributions

Better: first-order stochastic dominance

Definition · FOSD

F **first-order stochastically dominates** G if for every nondecreasing $u : \mathbb{R} \rightarrow \mathbb{R}$,
 $\int u(x) dF(x) \geq \int u(x) dG(x)$.

- Equivalent test: $F(x) \leq G(x)$ for all x — F puts more weight on high realizations, everywhere. And $\text{FOSD} \Rightarrow \int x dF \geq \int x dG$.
- Read it as: *anyone who prefers more money to less* prefers F , whatever their risk attitude. A strong result!

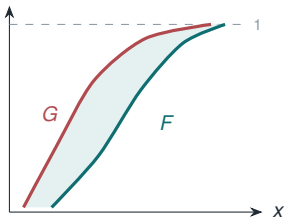
Riskier: second-order stochastic dominance

Definition · SOSD

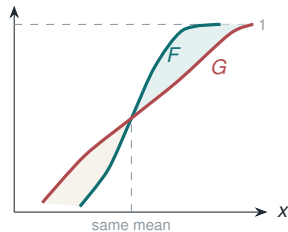
For F, G with the *same mean*: F **second-order stochastically dominates** G if for every nondecreasing *concave* u , $\int u(x) dF(x) \geq \int u(x) dG(x)$.

- G is a **mean-preserving spread** of F . Same mean, more dispersion.
- This is the formal content of "riskier": *every* risk-averse agent ranks F above G .
Where FOSD ordered *levels*, SOSD orders *risk* (variance) at fixed level.

In pictures



FOSD: $F(x) \leq G(x)$ everywhere — no crossing. F 's mass sits to the right: anyone who likes money prefers F .



SOSD: one crossing, *equal shaded areas* $\Leftrightarrow$ same mean. G moves mass to both tails — a mean-preserving spread: every risk-averter prefers F .

One more tool: updating the prior

- Expected utility runs on probabilities; where do they come from tomorrow? If beliefs are subjective probabilities, rationality extends: update by **Bayes' rule**,

$$\Pr(s_j | e) = \frac{\Pr(e | s_j) p_{s_j}}{\sum_h \Pr(e | s_h) p_{s_h}}.$$

- s = states of the world, p = prior, e = new evidence.
- Bayes says: We saw e , what is the chance e could have happened in each state? If it's high, it's now more likely we're in that state. Weighted by the overall chance of e .
- Our running prior — $\Pr(\text{shock}) = 0.2$ as of Dec. 2021 — is exactly what a household should have *revised* through 2022 as the draws arrived. Belief dynamics are lottery dynamics.

The ex ante cost of inflation

The setup: 2022 as a lottery, seen from December 2021

- Two states, prior $\Pr(\text{shock}) = 0.2$ (illustrative):

	normal ($p = 0.8$)	shock ($p = 0.2$)
prices	uniform $\pi = 2\%$	the 2022 draw ($g_E = 57.9$, $\pi_O = 6.3$)
bill, poor / rich	2.0% / 2.0%	11.3% / 9.3% (L6's full-price bills)

- NB the normal column: a uniform 2% costs every basket exactly 2%.
- Preferences: CRRA, with $r \in [1, 3]$. Each household's real income is a lottery; its **risk premium** is the ex ante cost of inflation risk.

Uses: expected utility; CRRA. Assumes: the two-state prior; previous results.

What would you pay ex-ante to not face the risk of 2022 inflation

risk premium of the uninsured lottery	$r = 1$	$r = 2$	$r = 3$
poor ($s_0 = 0.10$)	0.07%	0.15%	0.23%
rich ($s_0 = 0.06$)	0.05%	0.09%	0.14%

- **Ordering:** the poor's premium exceeds the rich's at every r — their lottery is the rich's, shifted down *and* spread out (bigger exposure to the relative-price draw).
- **Magnitude:** 0.1–0.6% of income — an order below the realized bill ($\sim 10\%$). What hurt was overwhelmingly the draw, not the uncertainty.

Proof → p. 25

The insurance value of indexation

The two lotteries a worker can hold

Previous calculations:

real income, by state	uninsured	indexed
normal ($p = 0.8$)	-2%	0
shock ($p = 0.2$)	-11.3 / -9.3%	-1.7 / +0.3%

Lotteries available:

		mean	variance (% ²)
poor	uninsured	-3.86%	13.84
	indexed	-0.34%	0.46
rich	uninsured	-3.46%	8.53
	indexed	+0.06%	0.01

- NB, at the moment ignore the job loss channel (+4.5pp under indexation).

The insurance value of indexation

insurance value ΔCE ($r = 2$)	total	mean part	risk part
poor	+3.67%	+3.52%	+0.15%
rich	+3.61%	+3.52%	+0.09%

- **The contract is valuable** — worth $\approx +3.6\%$ of income ex ante.
- Almost all of the value is the **mean** part: a state-contingent transfer that pays when times are bad. The pure risk-removal part is 0.1–0.15%.
- Unpriced: Unemployment risk.

When the assumptions are wrong

Failure I: independence (Allais) — what it does to our numbers

- Why it matters here: our results are certainty-equivalent numbers, and CE calculations require the linearity that Allais invalidates. If certainty is overweighted, the value of guaranteed indexation is higher than expected utility says — and fear of the uninsured tail is stronger.
- Allais-type behaviour raises the felt value of automatic indexation relative to our +3.2%; it does not overturn the decomposition.

Failure II: state-independent utility

- Utility is plausibly **state-dependent**: $u(x; \text{employed}) \neq u(x; \text{workless})$ — job loss changes health, time, status, not just the budget.
- If marginal utility is higher when workless, then: (i) the **job-hazard cost of indexation is larger**; (ii) symmetrically, transfers that pay in bad states are worth more — insurance equalises marginal utility across states.
- Direction: state-dependence **amplifies both sides** — the insurance value and the jobs price — and shifts the design question toward: insure the workless directly (benefits, already indexed) rather than only the employed's wage.

Failure III: objective probabilities

- We assumed the prior is known. Two failures.
- **Ellsberg**: offered bets on a known 50/50 urn vs an urn of unknown composition, most prefer the known — no single subjective probability rationalises the pattern.
Ambiguity aversion: people dislike not knowing the odds.
- **Measured inflation beliefs**: household inflation expectations are systematically biased upward and enormously dispersed, anchored on salient prices like groceries and fuel (survey evidence; Weber et al. 2022).
- Direction: if households cannot assess π -risk, they will not buy the right insurance privately. Ambiguity aversion + ambiguity over inflation risk $\Rightarrow$ insurance is worth more.

To index or not to index, that is the question

Should we index?

One shock, one policy, three margins:

margin	block	tool	the number
transfer accuracy, ex post	1	CV, Stone–Geary	bills 11.3/9.3 vs cheque 9.6 $\Rightarrow -1.7 / +0.3$
incidence, through firms	2	Shephard, CES	$+1.1$ pp prices, -4.5% jobs,
insurance, ex ante	3	EU, CRRA	risk 0.05–0.25%; mean $+3.3\%$

Roughly:

- Indexation undoes the € cost of inflation despite causing it to be a little worse.
- Average protection hides regressive heterogeneity.
- Cost of this protection is unemployment.
- Hidden benefit as an insurance is non-negligible.

What we missed

- The GE effects of higher real wages pushing up demand. **Pro-indexation.**
- Input-Output feedback of intermediate goods carrying the wage rise. **Anti-indexation.**
- Open-economy. Labor gets more expensive, domestic goods get more expensive, switch to imports. **Anti-indexation.**
- Heterogeneity is on many other dimensions than those we study. **Anti-indexation.**

What we assumed

- We made many assumptions to come to our answer.
- These will not necessarily be the right assumptions...
- They should be scrutinised, and the sensitivity of our conclusion to them quantified.
- Only because we were rigorous and formal can we clearly articulate our assumptions, and their empirical content, and therefore stress-test them.

Fin

Appendix

Derivations

The uninsured poor household's lottery over gross real income x : $x = 0.98$ w.p. 0.8, $x = 0.887$ w.p. 0.2. Four steps, at $r = 2$ (so $u(x) = -1/x$, $u^{-1}(v) = -1/v$):

1. **Mean:** $\mathbb{E}[x] = 0.8(0.98) + 0.2(0.887) = 0.9614$.
2. **Expected utility:** $\mathbb{E}[u] = -\left(\frac{0.8}{0.98} + \frac{0.2}{0.887}\right) = -(0.81633 + 0.22548) = -1.04181$.
3. **Invert:** $c = -1/\mathbb{E}[u] = 1/1.04181 = 0.95987$.
4. **Premium:** $\mathbb{E}[x] - c = 0.9614 - 0.95987 = 0.00153 \Rightarrow \mathbf{0.15\%}$ of expenditure.

Other cells, same recipe: $r = 1$ uses $\ln$: $\mathbb{E}[\ln x] = -0.04014$, $c = e^{-0.04014} = 0.96065 \Rightarrow 0.07\%$. $r = 3$ uses x^{-2} : $\mathbb{E}[x^{-2}] = 1.08719$, $c = 1.08719^{-1/2} = 0.95907 \Rightarrow 0.23\%$. Rich at $r = 2$: outcomes 0.98/0.907, $c = 0.96448 \Rightarrow 0.09\%$.

Derivations

Run the same four steps on the *indexed* lottery (poor: $x = 1.0$ w.p. 0.8, $x = 0.983$ w.p. 0.2), at $r = 2$:

$$\mathbb{E}[x] = 0.9966, \quad \mathbb{E}[u] = -\left(0.8 + \frac{0.2}{0.983}\right) = -1.00346, \quad c = 0.99655, \quad RP = 0.00005$$

1. **Total:** $\Delta CE = c_{\text{idx}} - c_{\text{unins}} = 0.99655 - 0.95987 = 0.03668 \Rightarrow +3.67\%$.

2. **Decompose** via $CE = \text{mean} - RP$, so

$$\Delta CE = \underbrace{0.9966 - 0.9614}_{= +3.52\text{pp}} + \underbrace{(RP_{\text{unins}} - RP_{\text{idx}})}_{0.15\% - 0.005\% = +0.15\text{pp}}$$

the “risk part” is the difference of the two premia: what variance-removal is worth.

3. **Rich**, same steps: $c_{\text{idx}} = 1.00060$, $c_{\text{unins}} = 0.96448$, $\Delta CE = +3.61\%$; mean part identical (+3.52: the cheque is common), risk part +0.09.